

Warming Amplifies the Frequency of Harmful Algal Blooms with Eutrophication in Chinese Coastal Waters

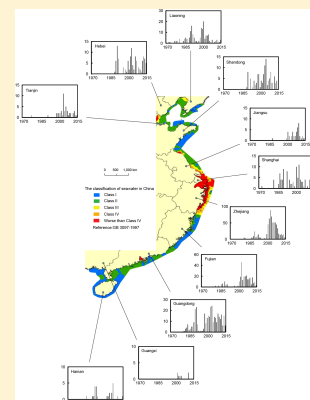
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Supporting Information

ABSTRACT: Widespread coastal eutrophication is known to increase the prevalence of harmful algal blooms (HABs). Increased HABs have also been linked to climate change, with ocean warming predicted to lead to increased prevalence and earlier timing of HABs. Testing the predictions of warming to HABs is difficult due to the lack of long-term observations across spatial scales. Here, we use a 45 year (1970–2015) record of the occurrence and duration of HABs along Chinese coast to show that the HAB frequency has increased at a rate of $40 \pm 4\%$ decade⁻¹, with earlier timing by 5.50 ± 1.78 days decade⁻¹. The increasing frequency of blooms varied with latitude and is significantly correlated with warming at an average rate of 0.17 ± 0.03 °C decade⁻¹, with the positive relationship being strongest in more eutrophic provinces. HAB frequency increased with elevated dissolved inorganic nutrient concentration, but this increase was amplified further with warming. Warming and eutrophication showed additive roles in triggering HABs. Swift action to mitigate eutrophication is essential to avoid a sharp increase in the HABs in coastal waters with further warming.



INTRODUCTION

Coastal ecosystems supply a major fraction of ecosystem benefits supporting human societies¹ concentrated in coastal areas.² In turn, high human population density and activity in coastal zones has led to elevated nutrient inputs, resulting in widespread eutrophication and increased harmful algal blooms (HABs) across the coastal ocean.^{3–5} Coastal HABs occur via production and accumulation (retention, physical transport, and behavior) of microalgae and may lead to major impacts on human health, economy, and coastal ecosystems.^{4,6,7}

Climate change has been hypothesized to lead to the expansion of blooming geographical domains and seasonal windows,⁸ and especially, more frequent and earlier blooms,^{5,7,9–15} a hypothesis consistent with reported trends in different regions.^{14,16–20} However, a paucity of long-term records across sufficiently large scales often precludes the clear attribution of alternations in HABs to climate change.²¹ For instance, the frequency of HABs tends to increase with nutrient inputs^{4,22} so that increasing eutrophication, a growing phenomenon along coastal waters, is also responsible for increasing HAB frequency^{4,22} and, therefore, confounded with the hypothesized response to warming. Moreover, attribution of any change in an ecosystem to climate change requires observational time scales in excess of 20–30 years and regional (100's of km) spatial scales.^{23,24} Such spatial scales are best covered by satellite observations ocean color, which were initiated in the 1980s¹⁷ so that assessments at regional scales predating satellite observations remain sparse. Further, accumulated CO₂ emissions and the consequent ocean

warming correlate with the increase in nutrient inputs to the coastal zone associated with a growing human population,^{8,19,25–28} making it difficult to separate the roles of climate change and eutrophication as important drivers of changes in coastal ecosystems. Moreover, climate change also affects a suite of processes, such as precipitation patterns, freshwater discharge, and vertical stratification,^{8,29–32} that can also affect algal blooms.^{16,18–20,33,34} For instance, in the rapid warming Indian Ocean, decreases in chlorophyll are driven by enhanced ocean stratification and suppressed nutrient mixing from subsurface layers in the past decades,³⁵ which have favored a shift toward green *Noctiluca*-based HABs in the Sea of Oman and Western Arabian Sea.^{36,37} Yet, despite model predictions of conditions that support the growth of HAB species with warming,¹⁴ the search for patterns in long-term observations of algal blooms has thus far been inconclusive, with no consistent patterns in changes in bloom frequency across coastal ecosystems.^{8,38} However, the responses of marine biota, including microalgae, to climate change also involve phenological alterations³⁹ that can be used as a fingerprint to separate climate-driven changes from eutrophication-driven changes.

Here, we examine long-term trends in the frequency and timing of HABs along the coast of China between 1970 and

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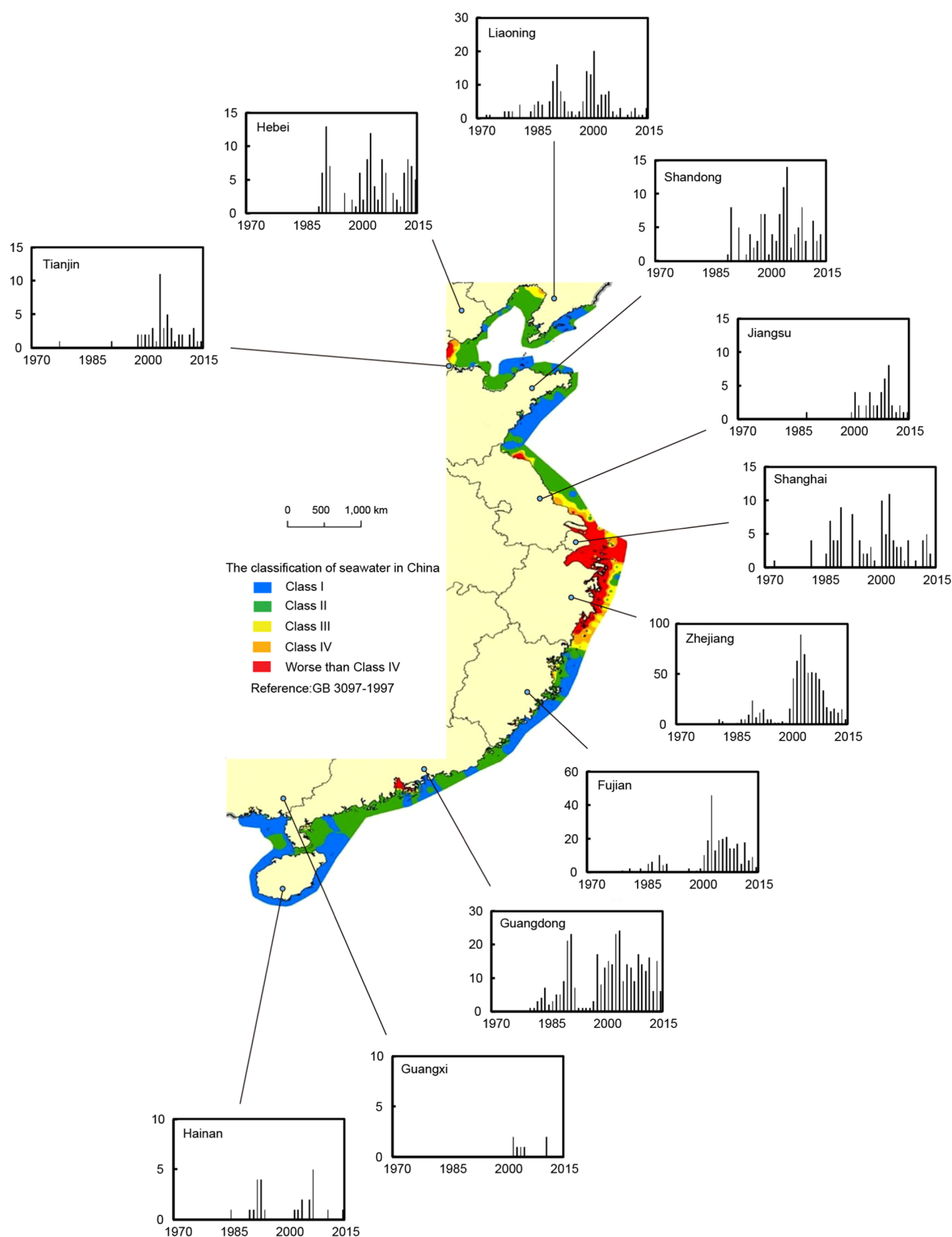


Figure 1. Coastal harmful algae bloom events and water quality along China's coastline. Surround: algal blooms recorded in Chinese coastal area from 1970 to 2015 for the 11 coastal provinces and municipalities. Middle: quality of coastal waters in 2015 classified into five classes from class I to class IV and worse than class IV according to the Chinese national standards (GB 3097-1997), modified from the China Environmental Status Bulletin. Red indicates worse status, and blue indicates good status in the water quality.

2015. This 45 year period encompasses a period of rapid economic development in China, resulting in widespread eutrophication and deterioration of coastal ecosystems.^{32,40–53} These impacts led to efforts to reduce nutrient inputs to alleviate coastal eutrophication through China's National Water Pollution Control and Treatment Science and Technology Plan initiated in 2008.⁵⁴

Although scientific reports of HAB events in the coastal areas of China date back to 1933,^{53,55} systematic reporting of HABs along China's coastal provinces has only taken place since 1964 when China's State Oceanic Administration (SOA) established, which developed into a national monitoring program managed by China's SOA.^{56,57} We compiled and synthesized here these hitherto fragmented records to examine the long-term frequency patterns of HABs along the coast of China and test the predicted relationship of changes in HAB frequency with long-term changes in sea-surface temperature⁵⁸ and dissolved inorganic nitrogen concentration, which is the major pollutant in Chinese coastal waters. We then examine seasonal patterns along the coast of China and test if these patterns shift, as predicted,^{14,15,59} with climate change.

MATERIALS AND METHODS

The database of algal blooms in China's coastal areas from 1933 to 2015 was acquired from public accessible data including (1) for the period 1933–2009 in two books,^{60,61} (2) for the period 2010 and 2015 derived from the annual reports from State Ocean Administration (SOA) and its branches, Ministry of Environmental Protection (MEP), and local governments of the 11 coastal administrative areas in China (as documented in Table S1). Because these annual reports overlap with the two summary books^{60,61} for some years, we confirmed that the data reported in the books corresponded to that reported in the annual reports.

Because of the importance of fisheries and aquaculture resources for China and the impacts of HABs on these resources, China has allocated significant efforts to monitoring these events. These efforts were initiated in 1933, but became systematic following the establishment of China's SOA in 1964. Methodology details and national standards for Marine Environmental monitoring in China are provided in the [Supporting Information](#).

Between 1970 and 2015, HAB detection was based on a combination of routine monitoring, contingent monitoring, and tracking monitoring, once HABs were detected. Routine monitoring used ship observations, supplemented with opportunistic reports to SOA of bloom sightings from aircrafts, satellites, and fishermen, subsequently confirmed by ship observations. The HABs monitoring guidelines are followed by all coastal administrative units (i.e., nine coastal provinces and two coastal municipalities; Figure 1). The monitoring standard (HY/T 069-2005) prescribes the frequency of monitoring to include routine monitoring, including four surveys per month during high-risk seasons and one to two surveys per month during other seasons, while contingent monitoring includes surveys every 3 days when harmful algal blooms were expected to occur, and tracking monitoring to be conducted daily when blooms were detected. The data reported include, for all blooms reported, the location and date of occurrence of the HAB, and, depending on sources or events, the duration of the HAB, the area covered and the shape of the area, the color of the water, the species involved, and the cell density of the dominant species.

We used all of these data sources to extract the location and date of occurrence (i.e., start) of HABs in Chinese waters from 1970 to 2015. We derived the longitude and latitude of the blooms from maps. The data set compiled included 1946 records of HABs in Chinese waters between 1970 and 2015 (Table S2, Figure 1). Whereas these data sources have been accessed for particular locations in a few papers^{53,62–64} and the methods were described in the paper by Wu et al.,⁶⁵ the data set has never been reported for the entire nation and broad time period as done here.

Time series of sea-surface temperature data (1970–2015) for the 11 coastal administrative areas were derived from the Met Office Marine Data Bank (Table S3) (<http://www.metoffice.gov.uk/hadobs/hadisst/data/download.html>).⁶⁶ Time series data of Dissolved Inorganic Nitrogen (DIN) for the 11 coastal administrative areas, the only nitrogen form reported in the Chinese monitoring standard, were derived from two sources, SOA (1979–2009, from the National Marine Environment Monitoring Centre, SOA) and MEP of China (2001–2015, from annual bulletins of environmental quality for coastal waters of China⁴⁰).

The examination of average trends across coastal China required normalization of the time series for each province to avoid giving higher weight to provinces with the highest HAB occurrence or the warmest waters. The standard score (*z*-score) was calculated to normalize the HABs frequency, which is a dimensionless quantity obtained by subtracting the mean HABs frequency from a frequency in individual region and then dividing the difference by the standard deviation. Specifically, the temporal trend in HABs across coastal China was represented as the average ($\pm$ SE) of the normalized (*z*-score) frequency of HABs across provinces. Likewise, the warming trend across coastal China from 1970 to 2015 and the dissolved inorganic nitrogen concentration were represented by the average normalized (*z*-scores) annual average seawater temperature and DIN across coastal China. Linear regression and nonparametric Kendall's test for monotonic trend were used in the time series analysis. For the Mann–Kendall–Tau test, the Sen's slope with two-sided intervals was estimated on the confidence level of 95% using Gilbert's modification of Theil/Sen method.

For the seasonality study, data on the timing of HABs were grouped into five coastal latitudinal bands, with each band encompassing 5° latitude (i.e., 38–43°N, 33–38°N, 28–33°N, 23–28°N, and 18–23°N). The northern (43–33°N), middle (33–28°N), and southern (18–23°N) areas correspond to warm-temperate, subtropical, and tropical climates, respectively. The total bloom frequency and the intermonthly variation of HABs were analyzed for each district in the modal month, i.e., the month with the highest frequency of HABs. The differences in the distributions of HAB frequency per week along the year between the five latitudinal bands was tested using the nonparametric Kolmogorov–Smirnov test.

For the temporal comparison, the HAB records for the recent decades spanning 1976–1985, 1986–1995, 1996–2005, and 2006–2015 were collected from the Chinese coastal HAB data set (CCHAB data set). We analyzed the temporal variation of Chinese coastal HABs on daily and weekly time scales. First, we calculated the rate of HAB frequency which is the frequency of HAB events on any specific day or week, thereby avoiding higher weight to decades with the highest HAB occurrence in the analysis if seasonality shifts. Pairwise comparisons of the distributions of HAB frequency per week

along the year between the four decades was tested using the nonparametric Kolmogorov–Smirnov test. Then, we identified the median/average/mode day and week with the median/mean/mode days/week of starting bloom days for each decade. Moreover, the slope of the linear regression between the modal day of occurrence with decade, encompassing the four decades from 1976 to 2015, provided an estimate of the temporal shift (as day/week per decade) in HAB seasonality.

Although the HABs monitoring program involves systematic assessments, we cannot rule out the possibility that changes in efforts over time may possibly confound long-term patterns in changes in HABs, as reported for coastal areas worldwide.⁷ We thus collected, as a proxy for effort, the number of persons engaged in marine environmental protection between 1981 and 2002 (with data gaps between 1993 and 1995) from the China Marine Statistical Yearbook.^{67,68}

RESULTS

In total, 1952 HAB events were reported along the 13 000 km of China's coast between 1933 and 2015. Of those, 1946 HABs occurred between 1970 and 2015 (Figure 1). During this period, 78.7% of China's coastal region experienced more than five HAB events, 31.9% experienced more than 50 events, and 12.8% experienced more than 100 HAB events (Figure S1). The occurrence of HABs was greatest in the Zhejiang and Guangdong provinces in southern China, accounting for 36 and 17% of the blooms, respectively (Figure 1). These two provinces are characterized by very poor coastal water quality (Figure 1). Indeed, there was a positive relationship between the mean HAB frequency and DIN concentration for the 11 coastal administrative areas in China, with the most nutrient-rich provinces having a higher prevalence of HABs (Figure S2). Furthermore, the frequency of HABs in China increased, on average, at a rate of $40 \pm 4\%$ decade⁻¹ (linear regression) and $39 \pm 8\%$ decade⁻¹ (Mann–Kendall–Tau test), between 1970 and 2015, with important oscillations around the increasing trend. The frequency of HABs in China reached a peak in 1990, when 84 HAB events were recorded, before declining to nine events in 1994. A second peak of 193 HAB events was recorded in 2003 before declining again to 29 events in 2015 (Figure 2a).

Chinese coastal waters warmed at an average rate of 0.17 ± 0.03 °C decade⁻¹ (linear regression) and 0.14 ± 0.10 decade⁻¹ (Mann–Kendall–Tau test) between 1970 and 2015 (Table S4). Seawater warming paralleled the increase in HAB frequency (Figure 2b), with warming accounting for 27% ($P < 0.001$) of the temporal variability in normalized HAB frequency in China over time (Figure 2b), which is a dimensionless quantity obtained by subtracting the mean HABs frequency from a frequency in individual regions and then dividing the difference by the standard deviation. The exponential rate of increase in HAB frequency with temperature was derived by fitting a linear regression between the natural log of the frequency of HABs and temperature. The slope of the fitted regression equation 0.555 ± 0.156 °C⁻¹ (Table S5) implies that the HAB frequency increased by, on average, $55.5 \pm 15.6\%$ with each degree celsius of warming. The positive relationship between HAB frequency and warming was strongest in the central provinces, where eutrophication was the most severe and HAB frequency highest (Figure 1), while this relationship was weakest in the northern provinces and in the southernmost Hainan province (Figure 3, Table S5). Indeed, provinces with high HAB

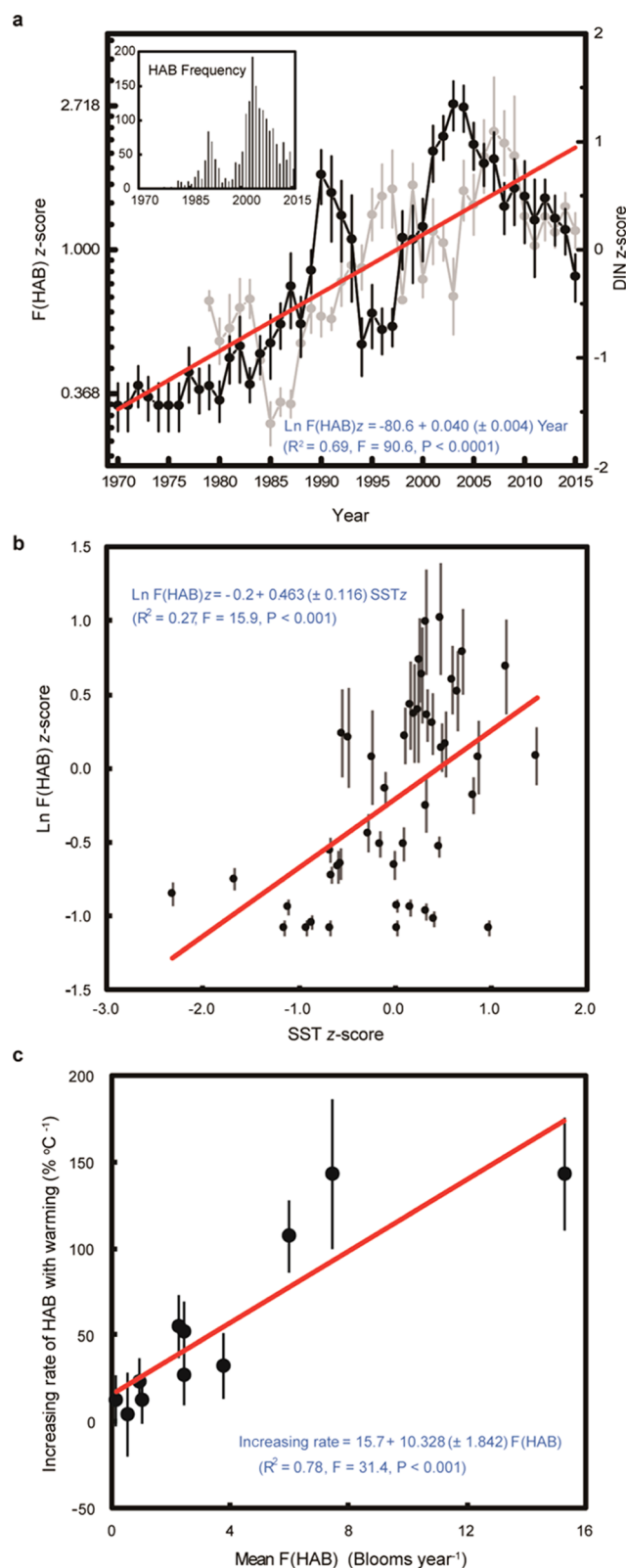


Figure 2. (a) Changes in the average ($\pm$ SE) normalized frequency of harmful algal blooms (black dots) and dissolved inorganic nitrogen (gray dots) across China from 1970 to 2015. Note the logarithmic axis for frequency of harmful algal blooms. The solid red line shows the fitted linear regression equation $\ln F(\text{HAB})z = -80.6 + 0.040 (\pm 0.004) \text{ year}$ ($R^2 = 0.69$, $F = 90.6$, $P < 0.0001$). The normalized frequency was calculated by normalizing each of the time series of the number of HAB events per year for each of the 11 provinces to z-scores (mean = 0, SD = 1) and then calculating the z-scores plus 1 (to

Figure 2. continued

avoid negative values) for each year across provinces. The inset shows the total number of HAB events detected over time. (b) The relationship between the annual average ($\pm$ SE) natural logarithm of the normalized (z-scores) frequency of harmful algal blooms, $\ln F$ (HAB) z , and the annual average normalized (z-scores) average seawater temperature across coastal China from 1970 to 2015. The solid red line shows the fitted equation $\ln F$ (HAB) $z = -0.2 + 0.463 (\pm 0.116)$ SST ($R^2 = 0.27$, $F = 15.9$, $P < 0.001$). The natural logarithm, $\ln F$ (HAB) z , was calculated by the natural log of the z-scores plus 1 (to avoid negative values) for each year across provinces. (c) The relationship between the average ($\pm$ SE) rate of increase in HAB frequency with warming and the average HAB frequency for the 11 coastal provinces of China from 1970 to 2015. The solid red line shows the fitted equation's increasing rate ($\%$, $^{\circ}\text{C}^{-1}$) = $15.7 + 10.328 (\pm 1.842)$ $F(\text{HAB})$ ($R^2 = 0.78$, $F = 31.4$, $P < 0.001$).

prevalence were more vulnerable to warming, as indicated by the lowest increase in HAB frequency in provinces with low HAB prevalence and larger increases in provinces with increasing HAB frequency ($R^2 = 0.78$, $P < 0.001$, Figure 2c).

Indeed, seawater temperature in Chinese coastal waters showed important fluctuations superimposed on the long-term warming trend (Figure S3). In addition to the climatic fluctuation, changes in nutrient concentration (Figure 2a) also add to fluctuations in HAB frequency, as the normalized (z-scores) average HAB frequency increased significantly with increasing normalized (z-scores) dissolved inorganic nitrogen (DIN) concentration ($P = 0.004$, Figure S2). Moreover, changes in the normalized (z-scores) seawater temperature and DIN concentration together accounted for 34% of the variance in normalized HAB frequency ($R^2 = 0.34$, $P = 0.0008$). However, analyses by province showed that, in general, changes in HABs within a province are independent of DIN for most of the coastal areas, except for Shandong and Jiangsu, for which a significant positive relationship between DIN and HAB frequency was found (Figure S4).

The seasonality of HABs also changed from north to south ($P < 0.01$), with a prevalent summer occurrence of HABs in the north shifting toward earlier occurrences toward the south, to reach the maximum frequency of occurrence in April in the southernmost region (Figure 4a). The seasonality of HABs also changed during the 45 year time series analyzed here, with a tendency toward an earlier median occurrence of maximum annual HAB frequency at a rate of 12.7 ± 2.8 days per decade earlier (Table 1). The distributions of HAB frequency per week along the year differed significantly among decades ($P < 0.01$), except for the oldest two decades (Figure 4; Tables S6 and S7).

There was no significant correlation between the frequency of HABs and the proxy for observational effort ($P > 0.05$), suggesting that our data on HAB frequency, resulting from systematic monitoring efforts, were not confounded by changes in effort over time (Figure S5).

DISCUSSION

Our results provide evidence that the frequency of HABs in China has increased over time concurrently with a shift toward earlier occurrence of HABs, as previously reported for the South Yellow Sea and East China Sea.^{32,41–52,69,70} Common species responsible for the HABs occurring in China included *Karenia mikimotoi*, *Phaeocystis globosa*, *Alexandrium* spp., *Prorocentrum donghaiense*, *Skeletonema costatum*, *Noctiluca*

scintillans, etc.,^{41,48,60,61,70} with HABs in the East China Sea dominated by *Prorocentrum donghaiense* and *Karenia mikimotoi*,^{42,52} and those in the South China Sea dominated by *Phaeocystis globosa*.⁴⁴ The frequency of HABs in China reached a peak in 1990, with 84 events recorded, and a second peak in 2003, with 193 events recorded, showing a clear basin-scale oscillation in the time period. Interestingly, marine organisms are often found to undergo such large oscillations, such as jellyfish populations with an approximate 20 years global oscillation.⁷¹ The important fluctuations around the long-term increasing trend of HABs frequency may be partly explained by climatic fluctuations, which have been shown to lead to large fluctuations in HAB frequency.⁷² The role of oceanographic oscillations, such as El Niño and other phenomena, on accounting for interannual oscillations in HABs frequency in China deserves further attention.^{63,73,74}

The role of climate change as a driver in the shifting the seasonality of HABs is supported by the significant shift in the distribution of HABs frequency along the year (Figure 4; Tables S6 and S7), and over time with the trend toward an earlier occurrence of HABs in the year by, on average ($\pm$ SE) 12.7 ± 2.8 days decade⁻¹ (Table 1). Overall, the frequency of HABs increased over time (Figure 2a) and the extent of temporal increases across provinces were dependent on DIN concentrations (Figure S2), which indeed helps explain the variability among regions. The seasonality shift experienced over time along Chinese coastal waters is consistent with reports of earlier seasonality of phytoplankton blooms in the Baltic Sea advancing by 6 days per decade,¹⁷ and, more broadly, the general tendency toward earlier phenological changes in marine organisms with ocean warming.³⁹ A similar analysis based on data with time scale of around 20 years might show different trends since the record is simply not long enough.

The role of ocean warming on the frequency of HABs is also demonstrated by the significant increase in HAB frequency along the Chinese coast at a rate, on average, of $55.5 \pm 15.6\%$ with each degree ($^{\circ}\text{C}$) of warming. The average rate of warming in Chinese coastal waters was 0.17 ± 0.03 $^{\circ}\text{C}$ decade⁻¹ (linear regression) and 0.14 ± 0.10 decade⁻¹ (Mann–Kendall–Tau test) between 1970 and 2015, which was twice the warming rate of 0.09 $^{\circ}\text{C}$ decade⁻¹ for the global ocean during this time,⁷⁵ consistent with warming rates derived from an analysis of large marine ecosystems in China (0.17 – 0.5 $^{\circ}\text{C}$ decade⁻¹).⁷⁶ Interestingly, the seasonal shift of HABs in China was much faster than the average shift of 6 days decade⁻¹ reported for Baltic Sea phytoplankton blooms and the rate of 4.4 ± 1.1 days decade⁻¹ reported for other marine organisms,³⁹ consistent with the much higher warming rate experienced along the coast of China compared to the average rate of warming in the global ocean.⁷⁵

The advance in seasonality over time does not, per se, demonstrate that this is driven by warming. However, advancing phenological shift is a globally coherent fingerprint of climate change,^{77,78} both on land^{79,80} and at sea.³⁹ Moreover, advancing phenology with climate change has been shown to apply to Chinese organisms on land.⁸¹ Provided warming of Chinese coastal waters reported here, and provided the compelling evidence pointing at advanced phenology as a fingerprint of climate change, it is possible to put forward the hypothesis that the advanced seasonality (i.e., timing of occurrence of HABs along the year) over time is indeed consistent with warming.

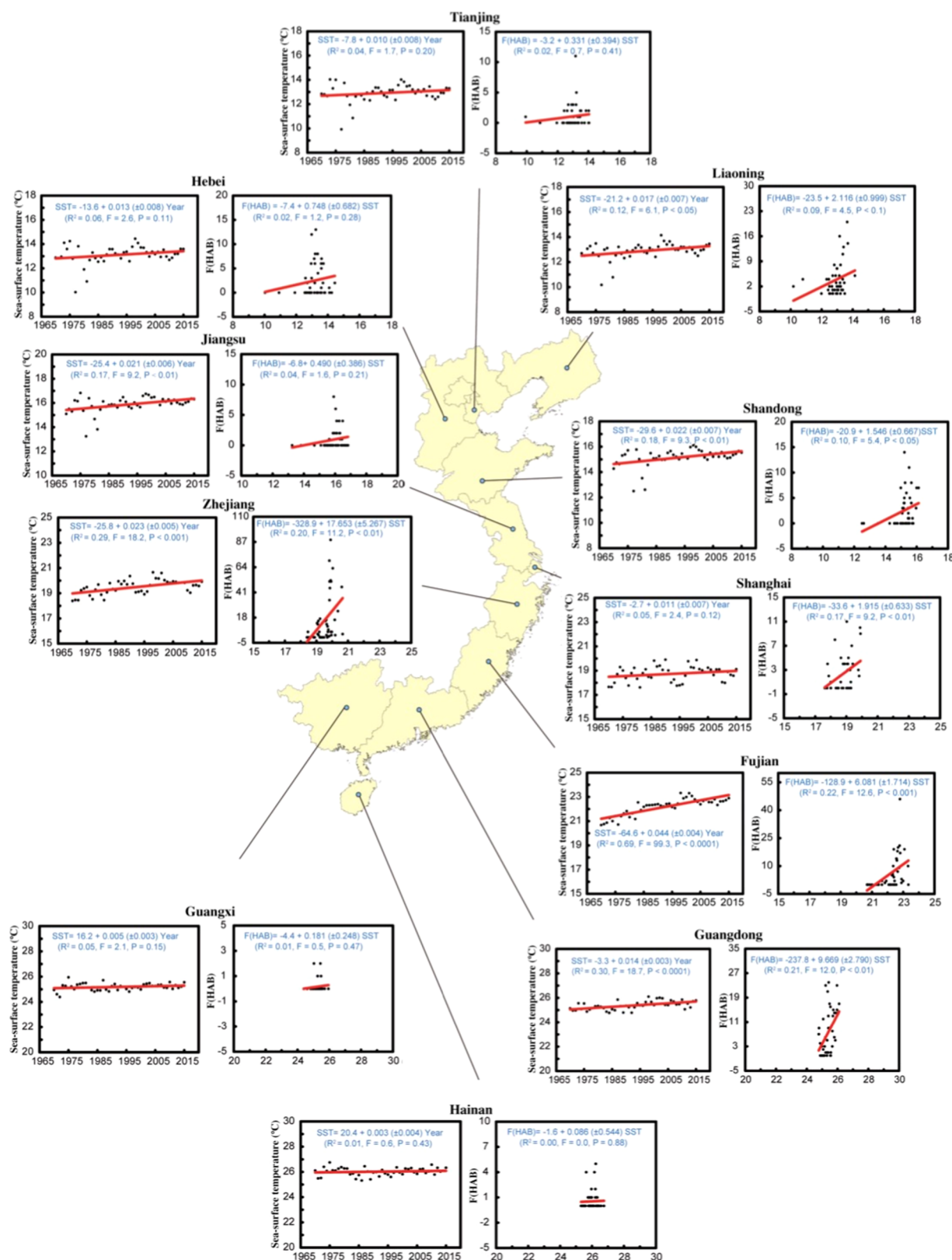


Figure 3. Sea-surface temperature in Chinese coastal area from 1970 to 2015 for the 11 coastal provinces and municipalities, and the relationship between the HAB frequency and sea-surface temperature.

The most eutrophic regions, which also supported the highest frequency of HABs, also experienced the highest rate of increase in HABs over time. The HABs and their temporal increase depended on DIN concentrations, thereby explaining

the variability of seasonal shift in the HABs among regions. This observation together with the increase in the frequency of HABs with increasing dissolved inorganic nutrient concentration for Chinese coastal waters reported here suggests that

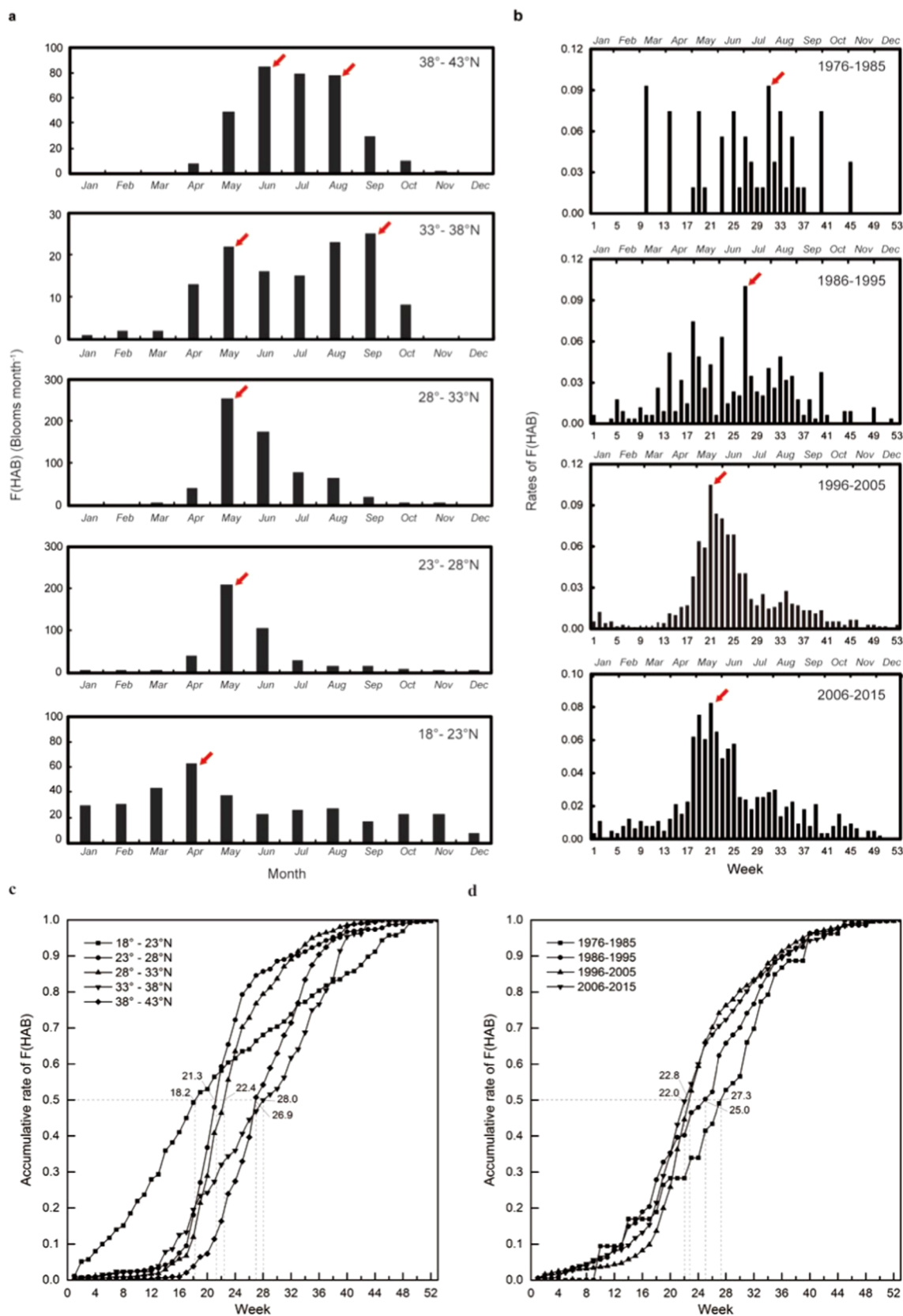


Figure 4. (a) Frequency histogram for intermonthly variation of coastal harmful algal blooms in China. The coastal harmful algal blooms from 1933 to 2015 were sorted according to the latitude of blooming locations, with an interval of 5°. The red arrow shows the month with the highest frequency of blooms. (b) Weekly rates of coastal harmful algal blooms in China for the recent four decades from 1976 to 2015. The red arrow shows the week with highest frequency of blooms. (c) The accumulative distributions of HAB frequency per week along the year for the five latitudinal bands. (d) The accumulative distributions of HAB frequency per week along the year for the recent four decades from 1976 to 2015.

Table 1. Changes in the Phenological Characteristics of Coastal HAB in China for the Recent Four Decades

	day ^a			week ^a		
	average	median	mode	average	median	mode
1976–1985	181	190	213	27	28	31
1986–1995	169	171	121	25	25	27
1996–2005	165	155	139	25	23	21
2006–2015	164	153	129	24	23	21
rates ^b (day/week per decade)	5.50 ± 1.78	12.70 ± 2.80	23.40 ± 16.23	0.90 ± 0.26	1.70 ± 0.48	3.60 ± 0.85

^aThe starting day or week of HAB events according to Julian calendar, from the Chinese coastal HAB database generated by this study. The median/average/mode day or week represents the day or week contains the median/mean/mode day of the blooms starting day list. ^bThe rates (day/week per decade) represent the average trend of seasonality toward advancement for the four decades from 1976 to 2015.

warming amplifies the risk of HABs with increasing nutrient concentration.^{59,82} Hence, global warming and coastal eutrophication act together to increase HAB frequency, suggesting that the likelihood of further increases in HABs with future warming is greatest in the most eutrophic coastal areas.⁸² Whereas we explore the relationship between HAB frequency and DIN, algal blooms can also be triggered by other nitrogen forms, including urea and organic nitrogen, and nutrients other than nitrogen, such as phosphorus.^{18,26,83,84} Unfortunately, data for other nitrogen forms are not reported in the Chinese monitoring standard and, accordingly, could not be evaluated here.

Our results are consistent with available regional reports of the occurrence and drivers of HABs in Chinese coastal waters, which report an increased frequency on HABs across Chinese coastal waters, along with an increase in the duration and extent of the blooms.^{44,70} These changes occur in relation to significant changes in biogeochemical conditions, with a shift from nitrogen to phosphorus limitation reported in the Changjiang River,⁵⁰ the NW Yellow Sea,⁴⁵ and Bohai Sea,^{42,43} as nitrogen inputs are dominated by atmospheric and nonpoint riverine sources, whereas phosphorus inputs are dominated by riverine inputs.^{43,45} Available reports also show a decline in silicate inputs due to reservoir construction and other changes in hydrology and sediment input, leading to a shift from diatoms to dinoflagellate dominance of HABs.^{42,43,45,70} In addition, increased irradiance due to decreasing sediment loads with the construction of the Three Gorge reservoir leads to increased phytoplankton blooms moving closer to the coast in the Changjiang Estuary.⁴⁹ The increase in dinoflagellate contribution to HABs in China has also been linked to increased vertical stratification and warming.^{45,70} The multiple concurrent changes in Chinese coastal waters, involving changes in nutrient stoichiometry, efforts to reduce nutrient inputs, hydrological changes brought by large-scale regulation of rivers with dam construction, together with climate change have been reported to curve the rising trend in HABs since year 2000 even if the eutrophication status of coastal waters has not been improved.⁷⁰

Our results support the hypothesized role played by ocean warming in the increase of the algal blooms frequency^{7,9,14,59} and, more specifically, in the frequency of HABs.^{5,10–13,15,58,59} Recently, niche analysis has predicted an increase in growth and niche expansion of HAB species with warming in the E. Pacific and N. Atlantic,¹⁴ consistent with the results presented here. Whereas our analysis refers to the frequency of HABs, a more detailed analysis of the monitoring program may also reveal changes in the community structure and the species responsible for these blooms over time and space. Such analysis will require information on the dominant species,

which is not yet available in our data set. No significant correlation showed up between the frequency of HABs and the proxy for observational effort. This is in agreement with reports by findings that HAB frequency in the South China Sea was independent of monitoring efforts and that increased monitoring efforts have only a small contribution to reports of increased HABs.⁶³ With increased dependence on the ocean and, specifically, on aquaculture, as a source of food, China, along with other nations vulnerable to HABs, must deploy strategies to mitigate the increased frequency of HABs in the future.⁸⁵ Our results suggest that these strategies require a reduction in nutrient input into coastal waters^{5,8,18–20,25–27,83,84} and leadership in reducing greenhouse gas emissions to the atmosphere,^{82,83} both of which are consistent with current policies in China.

In summary, whereas a rich body of literature had summarized the rise in HABs with changes in nutrients in coastal waters in different regions in China, as well as the curving of the rise in HABs since 2000,^{44,70} our analysis derives new insights into the role of ocean warming and its interaction with nutrient concentrations. Whereas previous analyses were based on observations at local to regional scales and narrower time windows, the data set contributed here provides a long-term record (45 years) consistent across the 11 coastal provinces of China, spanning from 18° to 43°N. This data set provides a solid foundation for our analysis, thereby allowing signatures of climate change, which had been previously suggested only,⁷⁰ to emerge. In particular, we demonstrate a delay in the seasonality of coastal HABs with increasing latitude along the Chinese coast, a relationship between HABs frequency and warming, also depending on nutrient concentrations, and a shift over time towards an advanced seasonality of HABs, which provide a fingerprint for the role of ocean warming in these changes. The demonstration that warming and eutrophication show additive roles in triggering HABs along with the patterns of change in the seasonality and frequency of HABs with warming provides novel understanding of the regulation of HABs in Chinese coastal waters that may help better manage and adapt to the impacts of this important problem.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.9b03726.

Methodology details and national standards for Marine Environmental monitoring in China; data sources and database of coastal HABs for China (1933–2015); annual mean sea-surface temperature (SST); results of

linear regressions and Mann–Kendall test of SST versus years and of the natural logarithm of HABs frequency versus SST; result of two-sample Kolmogorov–Smirnov test between the distribution of HABs frequency along the decades and latitude bands; results of Kendall Trend Test on the changes in the average ($\pm$ SE) normalized HABs frequency and SST; numbers of HABs events categorized on an interval of 0.5° ; relationship between the HABs frequency and the concentration of dissolved inorganic nitrogen (DIN); relationship between the normalized HABs frequency and the normalized DIN concentration; and Marine Environment monitoring efforts along years (PDF) (PDF)

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The authors declare no competing financial interest.

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